Naveen Shetty

Embedded Firmware Engineer

RESUME_1531

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Total Experience: 6 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 6 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: STM32, Git, SPI, C, FreeRTOS, CAN, Embedded Linux, ARM Cortex-M, JTAG, Firmware Debugging, C++, UART

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Senior Embedded Firmware Engineer | SparrowCloud Labs | Jan 2023 - Present

- Validated 618 hardware test vectors across protocol and timing corner cases
- Improved battery runtime by 15% with peripheral and sleep-state optimization
- Reduced embedded task jitter by 41% using interrupt priority and queue tuning
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Embedded Firmware Engineer | NovaPeak Engineering | Jul 2020 - Dec 2022

- Reduced embedded task jitter by 15% using interrupt priority and queue tuning
- Lowered field-reset incidents by 38% with watchdog recovery and diagnostics

PROJECTS

Motor Control Interface | ARM Cortex-M, STM32, UART, C++

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | FreeRTOS, C++, UART, Embedded Linux

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Silverlake School of Engineering, 2020
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

Embedded C Standards (2022)